

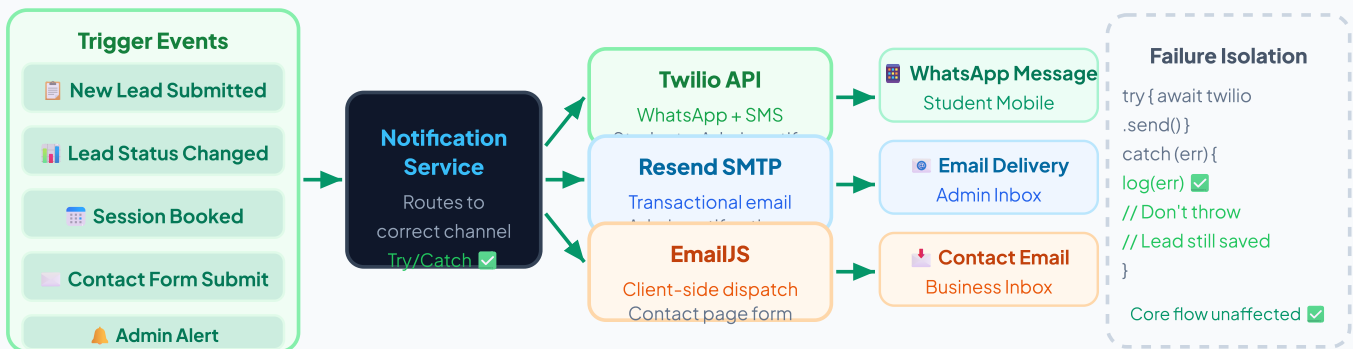
Communication & Messaging Integrations

Twilio WhatsApp/SMS gateway, Resend transactional SMTP, EmailJS client-side dispatch, SerpAPI live scholarship telemetry, and graceful failure isolation.

WhatsApp/SMS: Twilio REST Email: Resend SMTP + EmailJS Live Data: SerpAPI

1. Multi-Channel Communication Architecture

COMMUNICATION EVENT TRIGGERS — LEAD LIFECYCLE TO NOTIFICATION DISPATCH



2. Technology Decision Framework

1. Twilio WhatsApp vs Bulk SMS Services vs Native Messaging

MESSAGING

WHAT IS IT?

Twilio's REST API sends automated WhatsApp and SMS messages to students upon lead creation, status changes, and session bookings.

HOW DOES IT WORK & PROBLEM SOLVED?

Students in India have 90%+ WhatsApp penetration. Email open rates average 20%; WhatsApp open rates exceed 95%. When a lead is created, the notification service dispatches a WhatsApp message within 1-2 seconds, dramatically reducing lead response latency.

WHAT ALTERNATIVES EXISTED?

TextLocal / MSG91: India-specific SMS only, no WhatsApp Business API. **Direct WhatsApp API:** Requires Meta business verification (4-6 week process).

WHY DID WE CHOOSE THIS APPROACH?

Twilio provides a unified REST API for both WhatsApp and SMS fallback, with pre-approved templates for instant messaging and a single SDK for both channels.

2. Resend SMTP vs SendGrid vs AWS SES

EMAIL

WHAT IS IT?

Resend is a developer-first transactional email API built for modern web stacks with a simple `resend.emails.send()` API call.

HOW DOES IT WORK & PROBLEM SOLVED?

Admin notification emails (new lead alerts, conversion confirmations) are dispatched via Resend from the backend with high deliverability (authenticated SPF, DKIM, DMARC records managed by Resend's domain infrastructure).

WHAT ALTERNATIVES EXISTED?

SendGrid: Complex dashboard setup, older REST API. **AWS SES:** Excellent cost (\$0.0001/email) but requires manual SES production access approval and DNS configuration.

WHY DID WE CHOOSE THIS APPROACH?

Resend's SDK sends transactional email in 3 lines of code with SPF/DKIM handled automatically. Ship fast, high deliverability, free tier covers early-stage volume.

💡 Senior Interview Defense: "How do you handle notification delivery reliability?"

"UniCoach uses a three-channel communication strategy: Twilio for WhatsApp/SMS (95% open rate vs 20% for email), Resend for transactional admin emails, and EmailJS for client-side contact form dispatch. Each notification call is wrapped in a try-catch — if Twilio or Resend has an outage, the error is logged but the lead is still saved to MongoDB and the API returns 201. This failure isolation ensures a third-party messaging outage never breaks our core lead capture flow."